Business Requirements Document & Analytical Findings: Global Layoffs Analysis (2020-2023)

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1.0 Executive Summary

This analysis of global layoff data from March 2020 to March 2023 identifies two distinct economic events: the pandemic-driven shock of 2020 and a sustained downturn beginning in 2022. Key findings indicate that B2C industries (Consumer, Retail, Transportation) were disproportionately impacted, signaling reduced consumer spending. Furthermore, early-stage startups accounted for a minority of absolute job losses; the vast majority of layoffs were driven by large, post-IPO companies adjusting to economic pressures and market expectations. The findings are derived from a dataset with geographic bias toward U.S.-centric reporting.

2.0 Project Objectives & Scope

This project was started to improve skills with SQL by cleaning data and learning to create dashboards with Power BI. With the data, the objectives were to:

- Identify and analyze trends in workforce reductions.
- Pinpoint the most vulnerable industries and geographies.
- Assess the relationship between a company's funding stage and its layoff behavior.

Data Scope: The dataset comprises 2,361 layoff events from March 2020 to March 2023, with attributes including company, location, industry, total laid off, percentage laid off, date, stage, country, and funds raised.

3.0 Data Sources & Methodology

3.1 Data Sources

The dataset for this analysis was sourced from "Layoffs Dataset" on Kaggle. The dataset compiled reports from Bloomberg, The San Francisco Business Times, TechCrunch, and The New York Times.

3.2 Data Processing

The raw data was extracted, cleaned, and transformed using Power BI and SQL. Key preparation steps included:

- Handling missing/duplicate values and standardizing fields (e.g., standardized industries, removed companies with no layoff percentage and amount).
- Creating measures (e.g., cumulative layoffs for all startups and all established companies, total layoffs) using DAX..

3.3 Known Limitations & Biases

- **Geographic Bias:** The data sources are all U.S. media outlets. This results in a significant over-representation of U.S. companies, which must be considered when interpreting geographic distributions.
- **Reporting Bias:** The dataset captures layoff events that were deemed newsworthy by the source publications. It may under-represent layoffs in private companies or in regions with less media coverage.
- **Period Limitation:** The analysis is confined to March 2020 through March 2023.

4.0 Key Findings & Analytical Conclusions

4.1 Temporal Analysis: The Two Distinct Waves of Layoffs

- **Finding:** The data reveals two separate layoff cycles. The first was a local maximum in Q2 2020, consistent with initial COVID-19 lockdowns. The second began in February 2022, showing a sustained increase in layoffs, peaking in Q1 2023, the highest point in the three-year dataset.
- **Conclusion:** The market was not in a stable or healthy state at the dataset's conclusion. The 2022-2023 wave indicates a systemic economic downturn, distinct from the one-time pandemic shock, driven by factors such as rising inflation, recession fears, and market corrections in tech.

4.2 Industry Analysis: Consumer Sectors Bear the Brunt

- **Finding:** The Consumer, Retail, and Transportation industries were among the top five most affected sectors, making up a disproportionately high share of total layoff events.
- **Conclusion:** The concentration of layoffs in B2C industries is a strong indicator of reduced spending. This trend suggests underlying economic pressures on the average consumer, likely driven by inflation and recession fears, making these sectors the most vulnerable to layoffs.

4.3 Geographic Analysis: Acknowledging Data Source Bias

- **Finding:** The United States accounts for approximately 67% of all layoffs recorded, with other top countries including India, the Netherlands, Sweden, and Brazil.
- **Conclusion:** The significant skew in geographic distribution is a result of data sourcing from outlets like Bloomberg and The New York Times, which have a strong U.S. focus. Therefore, the data is acceptable for analyzing U.S. trends but is not suitable for

unbiased country-to-country comparisons. The "top countries" list is indicative of media coverage as much as economic reality.

4.4 Company Stage & Funding: Distinguishing Business Failure from Strategic Downsizing

- **Finding 1 (Scale of Job Loss):** Post-IPO companies were responsible for over half of all employees laid off (approx. 200,000), despite representing a smaller fraction of total layoff events. These companies also accounted for the majority of all funds raised in the dataset.
- **Finding 2 (Risk of Failure):** Startups (Seed, Series A-C) accounted for **63 total company shutdowns** (~54%), while established companies (Series D-Post-IPO) accounted for 20 (~17%). This indicates that a startup's failure rate is **over 3 times higher** than that of an established company.
- **Conclusion:** The data reveals a dual risk environment:
    1. **Startups face Existential Risk:** They are significantly more likely to fail, as shown by the overwhelming majority of 100% layoff events.
    2. **Established Companies Drive Absolute Job Loss:** Established companies account for a disproportionate number of job losses. However, the high volume of layoffs from well-funded, post-IPO firms does not signal failure. Instead, it shows large-scale cost-cutting in response to factors such as investor and public market pressure. Their massive employee base means that even small percentage cuts result in a large number of layoffs, dominating the overall figures.

5.0 Supporting Dashboards & Visualizations

The following interactive (if downloaded) Power BI dashboard was developed to visualize the findings outlined in this document. It allows users to explore the data on their own by focusing on specific stages, countries, or industries.

Global Layoff Analysis

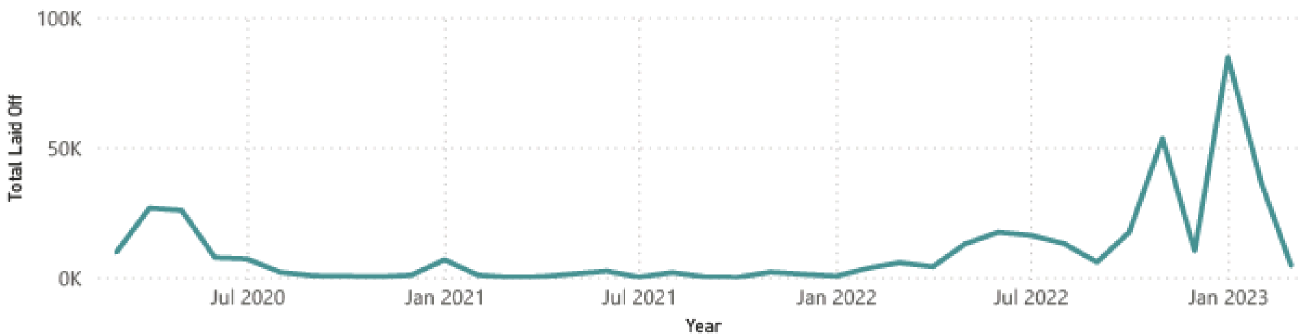
384K
Sum of Total Layoffs

31
Industries Affected

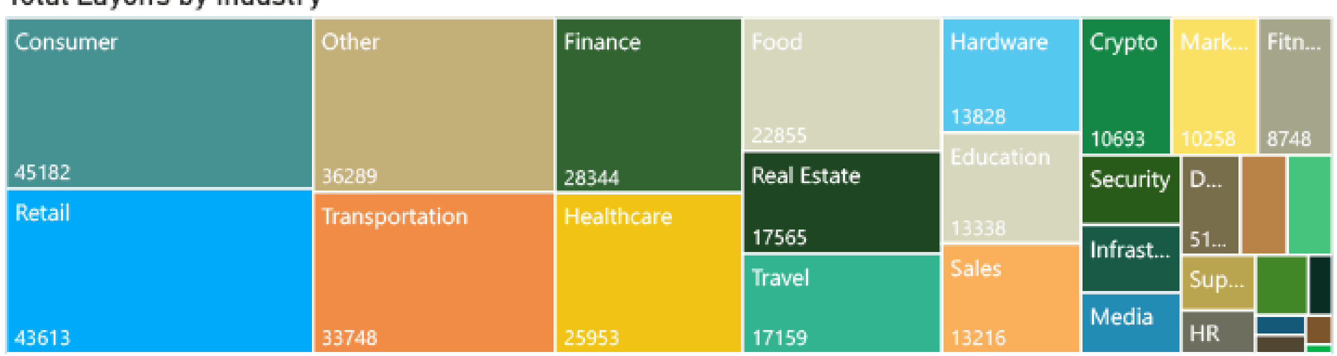
1995
Total Layoff Events

Jan 2023
Peak Month

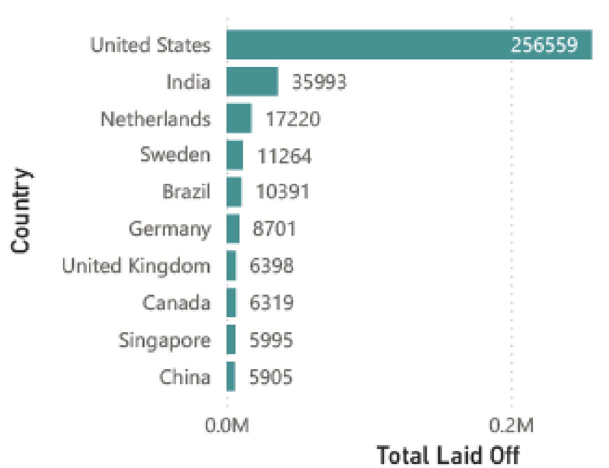
Total Layoffs per Month



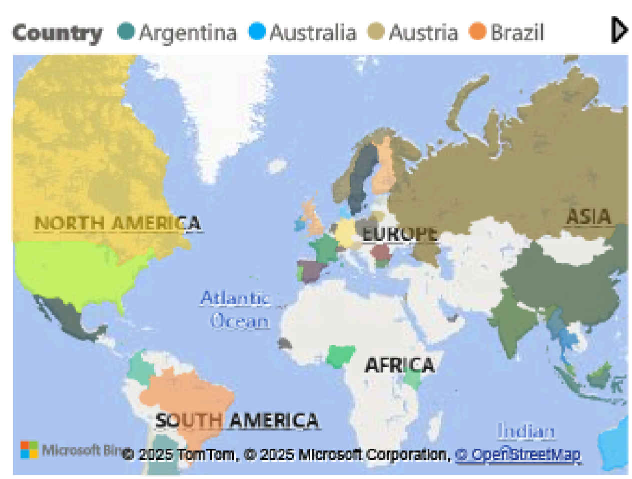
Total Layoffs by Industry



Top 10 Countries in Total Layoffs



Total Layoffs by Country

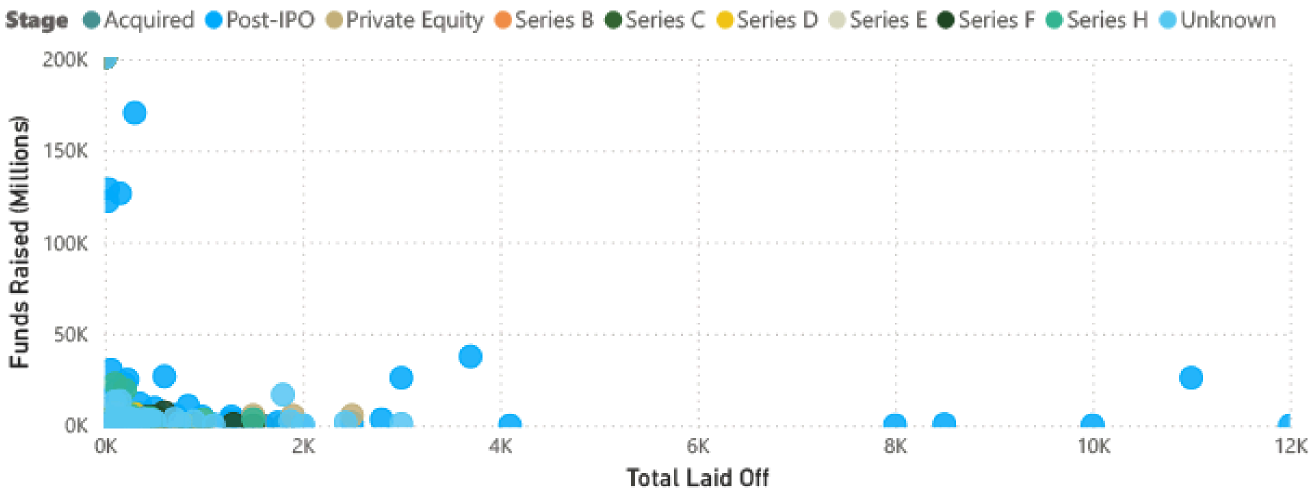


Company & Stage Analysis

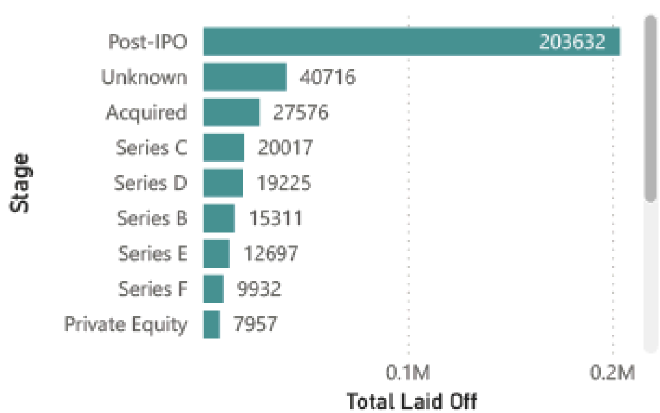
date

3/11/2020 — 3/6/2023

Layoffs vs Funding by Top 10 Stages



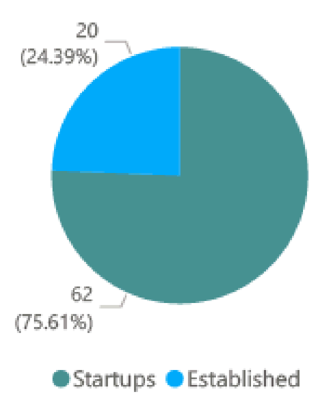
Layoffs by Stage



967
Established Layoff Events

707
Startup Layoff Events

Total Startups and Established Companies With 100% Layoffs



Funds Raised by Stage